```
    1
   0 1
  0 1 0
 1 0 1 0 1      n = 4
  1 0 1
   0 1
    1
```

First half.

intialization = $i = 1$; $j = 1$ ~~[illegible]~~ $n = 4$

Loop 1 Condition = $j \le n-i$; $i \le n$ → for spaces.

when

$i = 1$ j will become = $4-1 = 3$

$i = 1 \le 5$ ✓ $j = 1 \le 3$ ✓ print (" ")

$i = 1 \le 5$ ✓ $j = 2 \le 3$ ✓ print (" ")

$i = 1 \le 5$ ✓ $j = 3 \le 3$ ✓ print (" ")

$i = 1 \le 5$ ✓ $j = 4 \le 3$ ✗ —

when $i = 2$ j will become $4-2 = 2$

$i = 2 \le 5$ ✓ $j = 1 \le 2$ ✓ print (" ")

$i = 2 \le 5$ ✓ $j = 2 \le 2$ ✓ print (" ")

$i = 2 \le 5$ ✓ $j = 3 \le 2$ ✗ —

⋮

Same till $j = 1$

i

Loop 2 Condition $i \le n$ $j \le i$

~~i=1 ≤5 ✓ ; j~~

$i = 1 \le 5$ ✓ ; $j = 1 \le 1$ ✓
$i = 2 \le 5$ ✓ ; $j = 2 \le 1$ ✗

$i = 2 \le 5$ ✓ ; $j = 1 \le 2$ ✓
$i = 2 \le 5$ ✓ ; $j = 2 \le 2$ ✓
$i = 2 \le 5$ ✗ ; $j = 3 \le 2$ ✗
⋮ ⋮ ⋮ ⋮ ⋮

Similar till $i = 5$

$i = 6 \le 5$ ✗

Print Statement: if (count = True) print (0)
else print (1)
count == False.

$i = 1$; $j = 1$ count = False print → 1.

$i = 2$; $j = 1$ count = true print → 0
$i = 2$; $j = 2$ count = False print → 1

$i = 3$; $j = 1$ count = True print → 0
$i = 3$; $j = 2$ count = False print → 1
$i = 3$; $j = 3$ count = True print → 0
⋮ ⋮ ⋮ ⋮

Similar till $i = 5$

For Second Part also Similar just reverse (i)

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```
1           1
   2     2
      3
   4     4
5           5
```

intialization : $i = 1$; $j = 1$

Condition ; $i \le n$; $j \le n$; $n = 5$

iteration = 1

$i = 1 \le 5$ ✓ ; $j = 1$ $1 \le 5$ ✓
$i = 1 \le 5$ ✓ ; $j = 2$ $2 \le 5$ ✓
$i = 1 \le 5$ ✓ ; $j = 3$ $3 \le 5$ ✓
$i = 1 \le 5$ ✓ ; $j = 4$ $4 \le 5$ ✓
$i = 1 \le 5$ ✓ ; $j = 5$ $5 \le 5$ ✓
$i = 1 \le 5$ ✓ ; $j = 6$ $6 \le 5$ ✗

print j → if $(i = j) \,||\, i + j = n + 1$

$i = 1 \le 5$ ✓ ; $j = 1$ $i = j$ ✓ || $i + j = 6$ ✗ print (i)
$i = 1 \le 5$ ✓ ; $j = 2$ $i = j$ ✗ || $i + j = 6$ ✗ —
$i = 1 \le 5$ ✓ ; $j = 3$ $i = j$ ✗ || $i + j = 6$ ✗ —
$i = 1 \le 5$ ✓ ; $j = 4$ $i = j$ ✗ || $i + j = 6$ ✗ —
$i = 1 \le 5$ ✓ ; $j = 5$ $i = j$ ✗ || $i + j = 6$ ✓ print (i)
⋮

Similar for iteration 2 3 4

when $i = 6$
$i \le 6$ ✗ — — — — — —

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Double Number pyramid.

First half : initalization - $i = 1$; $j = 1$, $n = 4$.

(loop 1) Condition - $j \le n - i$; $i \le n$ → for spaces

When $i = 1$; j will become $4 - 1 = 3$

$i = 1 \le 5$ ✓ $j = 1 \le 3$ ✓ print (" ")
$i = 1 \le 5$ ✓ $j = 2 \le 3$ ✓ print (" ")
$i = 1 \le 5$ ✓ $j = 3 \le 3$ ✓ print (" ")
$i = 1 \le 5$ ✓ $j = 4 \le 3$ X —

When $i = 2$ j will become $4 - 2 = 2$

$i = 1 \le 5$ ✓ $j = 1 \le 2$ ✓ print (" ")
$i = 2 \le 5$ ✓ $j = 2 \le 2$ ✓ print (" ")
$i = 2 \le 5$ ✓ $j = 3 \le 2$ X —
⋮ ⋮ ⋮

Same till $j = 1$

(loop 2) Condition $i \le n$ $j \le i$

$i = 1 \le 5$ ✓ ; $j = 1 \le 1$ ✓ print (j)
$i = 2 \le 5$ ✓ ; $j = 2 \le 1$ X ~~print (j)~~ —
$i = 2 \le 5$ ✓ ; $j = 1 \le 2$ ✓ print (j)
$i = 2 \le 5$ ✓ ; $j = 2 \le 2$ ✓ print (j)
$i = 2 \le 5$ X ; $j = 3 \le 2$ X —
⋮ ⋮ ⋮ ⋮

Similar till $i = 5$
$i = 6 \le 5$ X - - - - - -

For Second Part similar just reverse ①

Number Butterfly.

initialization : $i \le n$ $j = 1$ $n = 4$

inner Loop = 1

Condition $i \le n$; $j <= i$; j

$i \le 1$; $j \le 1 \le 1$ ✓ ; Print ("j")
$i = 1$; $j = 2 \le 1$ ✗ ; —

$i \le 2$; $j \le 1 \le 2$ ✓ ; print (j)
$i \le 2$; $j \le 2 \le 2$ ✓ ; Print (j)
$i \le 2$; $j = 3 \le 2$ ✗ ; —

$i = 3$; $j = 1 \le 3$ ✓ ; print (j)
$i = 3$; $j = 2 \le 3$ ✓ ; print (j)
$i = 3$; $j = 3 \le 3$ ✓ ; print (j)
$i = 3$; $j = 4 \le 3$ ✗ ;
⋮

Same till $\boxed{i = 5}$

Loop ②

condition $i \le n$ $K = 2*(n-i) = 2*(4-i)$

when $i = 1$ $j = 2*(4-1) = 2*(3) = 6$

$i = 1 \le 4$ ✓ ; $j = 1 \le 6$ ✓ Print (" ") ;
$I = 1 \le 4$ ✓ ; $j = 2 \le 6$ ✓ Print (" ") ;
$i = 1 \le 4$ ✓ ; $j = 3 \le 6$ ✓ Print (" ") ;
$i = 1 \le 4$ ✓ ; $j = 4 \le 6$ ✓ Print (" ") ;
$i = 1 \le 4$ ✓ ; $j = 5 \le 6$ ✓ Print (" ") ;
$i = 1 \le 4$ ✓ ; $j = 6 \le 6$ ✓ Print (" ") ;
$i = 1 \le 4$ ✓ ; $j = 7 \le 6$ ✗ —

Loop 3 ;

Condition $i \le n$ $j \le i$

$i = 1$; ≤ 4 ✓ $j = 1$; ≤ 1 ✓ print (j)
$i = 1$; ≤ 4 ✓ $j = 2$; ≤ 2 X ~~print (j)~~

$i = 2$; ≤ 4 ✓ ; $j = 1$ ≤ 2 ✓ print (j)
$i = 2$; ≤ 4 ✓ ; $j = 2$ ≤ 2 ✓ print (j)
$i = 2$; ≤ 4 X ; $j = 3$ ≤ 2 X —

$i = 3$ ≤ 4 ✓ ; $j = 1$ ≤ 3 ✓ print (j)
$i = 3$ ≤ 4 ✓ ; $j = 2$ ≤ 3 ✓ print (j)
$i = 3$ ≤ 4 ✓ ; $j = 3$ ≤ 3 ✓ print (j)
$i = 3$ ≤ 4 X ; $j = 4$ ≤ 3 X —

⋮ ⋮ ⋮ ⋮ ⋮

Similar till $i = 4$

Part - 2

Similar but reverse the (i) loop

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4. Spiral layer:

here if $n = 3$ then row will be $(2*n)-1$
$\therefore$ for $n=3$; $(2*3)-1 = 6-1 = 5$

Condition: $i \le n$; $j \le$ row row $= 5$

$i=1 \le 5$ ✓ ; $j \le 1$ $1 \le 5$ ✓
$i=1 \le 5$ ✓ ; $j \le 2$ $2 \le 5$ ✓
$i=1 \le 5$ ✓ ; $j \le 3$ $3 \le 5$ ✓
$i=1 \le 5$ ✓ ; $j = 4$ $4 \le 5$ ✓
$i=1 \le 5$ ✓ ; $j = 5$ $5 \le 5$ ✓
$i=1 \le 5$ ✓ ; $j = 6$ $6 \le 5$ ✗

Print: first find distance.

① $i=1$ $j=1$.

int top $= i-1 = 1-1 = 0$
int bottom $=$ row $- i = 5-1 = 4$
int left $= j-1 = 1-1 = 0$
int right $=$ row $- j = 5-1 = 4$

Min among this is 0 then layer no $= 0$

therefore print $(n -$ layer no$) \Rightarrow (3-0) = 3$

in $(1,1)$ 3 will be printed:

Similar till $i = 5$

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* Pascal's triangle :-

intialization $i = 0$; $j = 0$; $n = 5$

loop 1:

Condition: $i = n$; $j \le n - i$

$i = 0$; $j = 5 - 0 = 5$

$i = 0 \le 5$ ✓ ; $j = 0$ ✓ ; print (" ")
$i = 0 \le 5$ ✓ ; $j = 1$ ✓ ; print (" ")
$i = 0 \le 5$ ✓ ; $j = 2$ ✓ ; print (" ")
$i = 0 \le 5$ ✓ ; $j = 3$ ✓ ; print (" ")
$i = 0 \le 5$ ✓ ; $j = 4$ ✓ ; print (" ")
$i = 0 \le 5$ ✓ ; $j = 5$ ✓ ; print (" ")
$i = 0 \le 5$ ✓ ; $j = 6$ ✗ ; — — —
⋮ ⋮ ⋮

Similar $i = 5$

when $i = 6$; $j = 5 - 6 = -1$

$i = 6 \le 5$ ✗ — — — — — —

loop 2:

condition ; $i = 0$; $k = 0$;

condition ; $i \le 5$; $j \le i$

$i = 0 \le 5$ ✓ ; $j = 0 \le 0$ ✓ print (value)
$i = 0 \le 5$ ✓ ; $j = 1 \le 1$ ✗ —

$i = 1 \le 5$ ✓ ; $j = 0 \le 1$ ✓ print (value)
$i = 1 \le 5$ ✓ ; $j = 1 \le 1$ ✓ print (value)
$i = 1 \le 5$ ✓ ; $j = 2 \le 1$ ✗ —

$i=2$	≤ 5 ✓	;	$j=0$	≤ 2	✓	print (value)
$i=2$	≤ 5 ✓	;	$j=1$	≤ 2	✓	print (value)
$i=2$	≤ 5 ✓	;	$j=2$	≤ 2	✓	print (value)
$i=2$	≤ 5 ✓	;	$j=3$	≤ 2	X	— —
⋮	⋮		⋮	⋮	⋮	⋮

Similar for $i=3$ $i=4$ $i=5$

when $i=6$

$i=6 \le 5$ X — — — — —

(" ") print statement.

$i=0; j=0;$ value $=1$; value $= 1 * (0-0/0+1) = 1$
$i=0; j=1;$ value $=1$; value $= 1 * (0-1/1+1) = 0$
$i=0; j=2;$ value $=-1$; value $= -1 * (0-2/2+1) = 0$
$i=0; j=3;$ value $=1$; value $= 1 * (0-3/3+1) = 0$
⋮ ⋮ ⋮ ⋮ ⋮ ⋮

Similar for $i=2$; $i=3$, $i=4$; $i=5$